Cover Page

- **Problem Statement/Concept Title**

  ESP32-CAM Based Smart Surveillance & Voice Controlled Robot

- **Theme**

  AI, GIS & Emerging Technologies

- **Team Name (Registered on portal)**

  TECH Warriors

- **Team ID (Registered on portal)**

  KARMA-X

- **State/UT**

  Uttar Pradesh

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Problem – Proposed Solution

Core problem / Proposed Solution

Affordable mobile monitoring is limited by fixed surveillance, lack of mobility, and the cost of commercial robotic platforms.

Proposed solution

- ESP32-CAM provides Wi-Fi communication and live camera streaming.
- L298N controls the 4-wheel drive; HC-SR04 supports obstacle detection.
- OLED displays status, calculator results and information.
- Voice control is being developed for forward, backward, left, right and stop.
- Three SG90 servos provide adjustable camera positioning; rechargeable power keeps the platform mobile.

Why it is unique

- Camera + mobility
- Voice + wireless control
- Obstacle sensing
- OLED interaction
- Low-cost modular platform

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Technical Approach

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Technologies

ESP32-CAM • L298N • 4 DC gear motors • SH1106 OLED • HC-SR04 • 3× SG90 • INMP441 • PAM8403 + speaker • 2×18650 + 2S BMS + 8.4V charger

Implementation flow

Rechargeable Power → **ESP32-CAM Controller** → **Wi-Fi / Voice Commands** → **L298N + 4WD Motors** → **Mobile Robot**

HC-SR04 → obstacle detection | OLED → status/calculator monitoring | SG90 → camera position | Camera → live

GitHub/GitLab: To be added • Demo Video: To be added • AI-tool usage: To be documented by team

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Impact Potential

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Potential impact on target audience

- Students and educational institutions: practical robotics, IoT and embedded-system learning.
- Small laboratories and project teams: affordable experimentation and modular prototyping.
- Basic monitoring applications: mobile camera-based observation where mobility is useful.
- Future developers: expandable base for sensors, automation and AI features.

Benefits

- Low-cost and rechargeable
- Combines monitoring, control, sensing and information display
- Modular architecture supports future upgrades

Stakeholders

- Students / colleges
- Project teams / labs
- Basic monitoring users
- Future robotics developers

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Feasibility and Risk Analysis

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Feasibility

- Uses widely available embedded modules and a modular 4WD platform.
- Subsystems can be tested independently before final integration.
- Prototype is rechargeable and designed for student-level implementation.

Potential challenges / risks

- Integrating camera, voice, motors and sensors on one controller.
- Reliable voice-command processing and live monitoring.
- Power, battery, BMS and charging integration.
- Stable camera positioning and obstacle detection.

Strategies

- Develop and test module-by-module.
- Use protected rechargeable power architecture.
- Validate movement and sensing before full integration.
- Improve voice, vision and navigation incrementally.

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Team

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Team Composition

Kartikey Kumar

Documentation, Hardware and Power control

Arpit Vishwakarma

Software, Hardware and Power control

Mayank Sahu

Coding and Testing

Challenges faced

- Integrating multiple modules into one compact platform.
- Testing each subsystem before final integration.
- Balancing mobility, monitoring, sensing and power requirements.

Team journey: Built the prototype module-by-module, validating ESP32-CAM and OLED functions before expanding to mobility, sensing and voice control.

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References

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References / Research Basis

- Smart Robot Concept Document — provided project concept source.
- HSC Idea Submission – Template v7.0 — provided presentation structure and submission pointers.
- Component/module documentation and datasheets to be added by the team as implementation references.

Note: The presentation content is based on the uploaded concept document and adapted to the supplied HSC template.